

sdstudio

A Companion Package for Designing
and Managing surveydown Surveys

Pingfan Hu

George Washington University

GW OSCON 2026



About Me



- **Pingfan Hu**, PhD Candidate at George Washington University, advised by Dr. John Helveston.
- Research interests:
 1. **Data Science** in sustainable transportation and human behavior
 2. **Open-Source Software** in **R** tools for survey research
 3. **Agentic AI Engineering** around Claude Code
- For more information, visit pingfanhu.com

jhelvy.shinyapps.io



Welcome to the Ultimate Superhero Survey!



Thank you for participating in our research on superhero preferences and powers! This survey will help us understand what makes superheroes appealing to different people.

This survey should take about 3-5 minutes to complete.

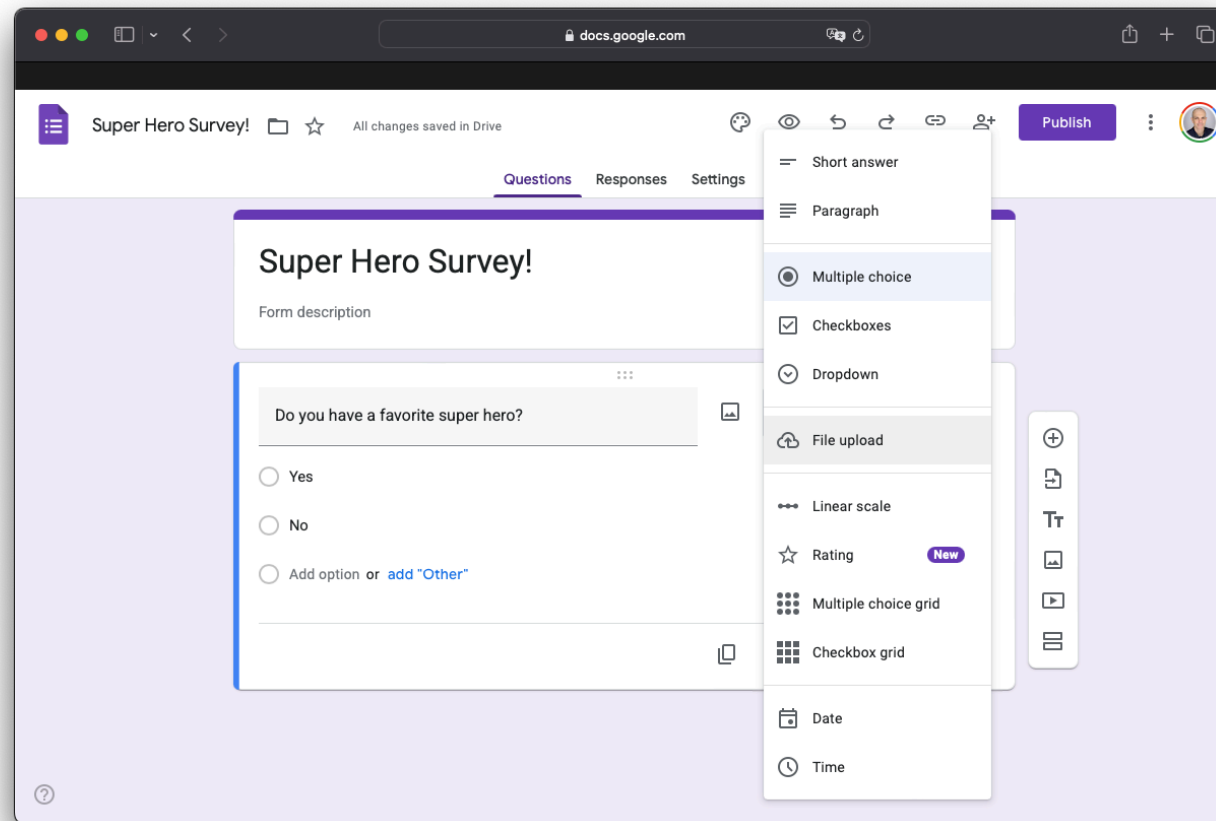
Do you have a favorite super hero?*

☐ Yes

☒ No

Next

WYSIWYG Interface



- ⊗ Reproducibility
- ⊗ Version control
- ⊗ Limited features
- ⊗ Open source

Why not make surveys
from **code**?

~~Limitations~~

Features

- ✓ Reproducibility
- ✓ Version control
- ✓ Lots of features
- ✓ Open source

Introducing **surveydown**!



Survey design using Quarto and Shiny



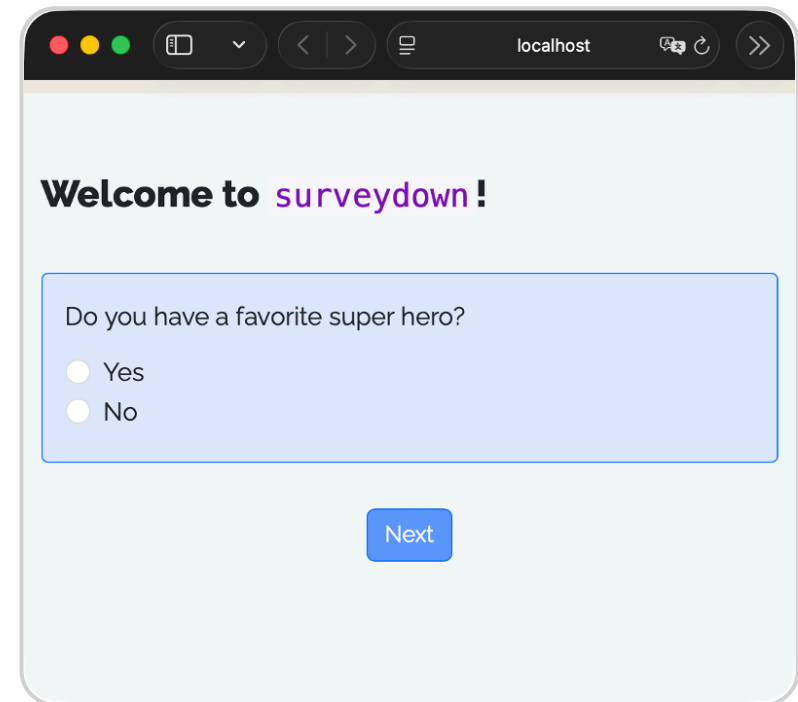
+



+



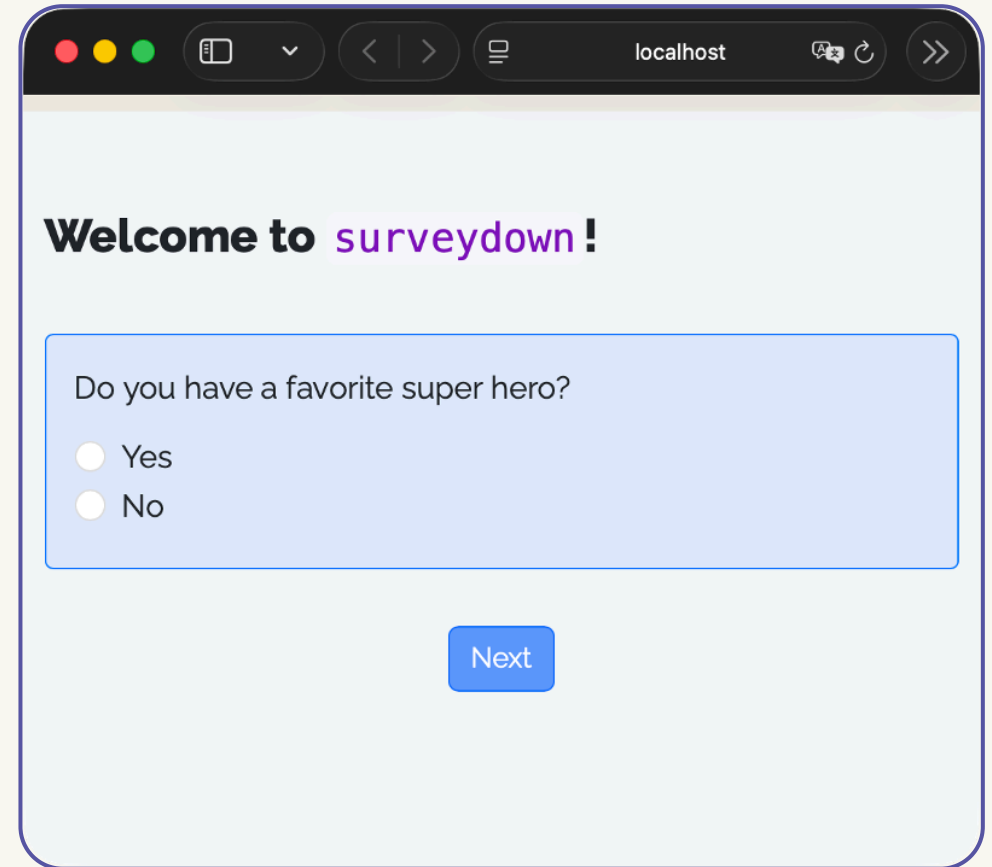
=



survey.qmd

```
1 ---
2 survey-settings:
3   use-cookies: true
4   rate-survey: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 --- welcome
12
13 # Welcome to `surveydown`!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id    = "has_fav_hero",
19   label = "Do you have a favorite superhero?",
20   option = c(
21     "Yes" = "yes",
22     "No"  = "no"
23   )
24 )
25 ```
26
27 --- some_other_page
28
29 Other content...
```

Rendered Survey



localhost

Welcome to surveydown !

Do you have a favorite super hero?

☐ Yes

☐ No

Next

But...



Code-Only Interface

No GUI. Not everyone is comfortable working purely in code.



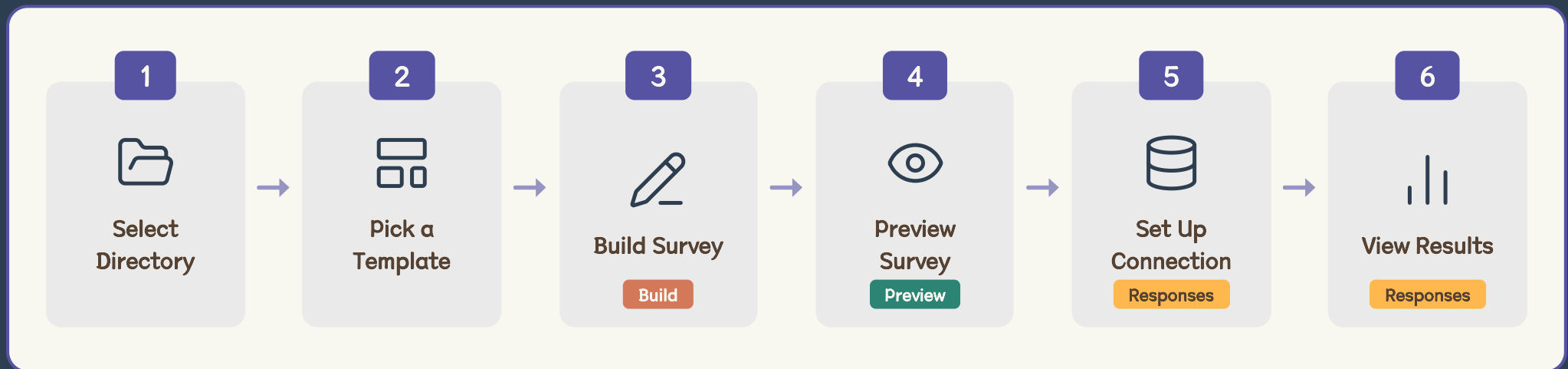
Learning Curve

The survey pipeline is not obvious. It takes time to learn.

Introducing **sdstudio**!



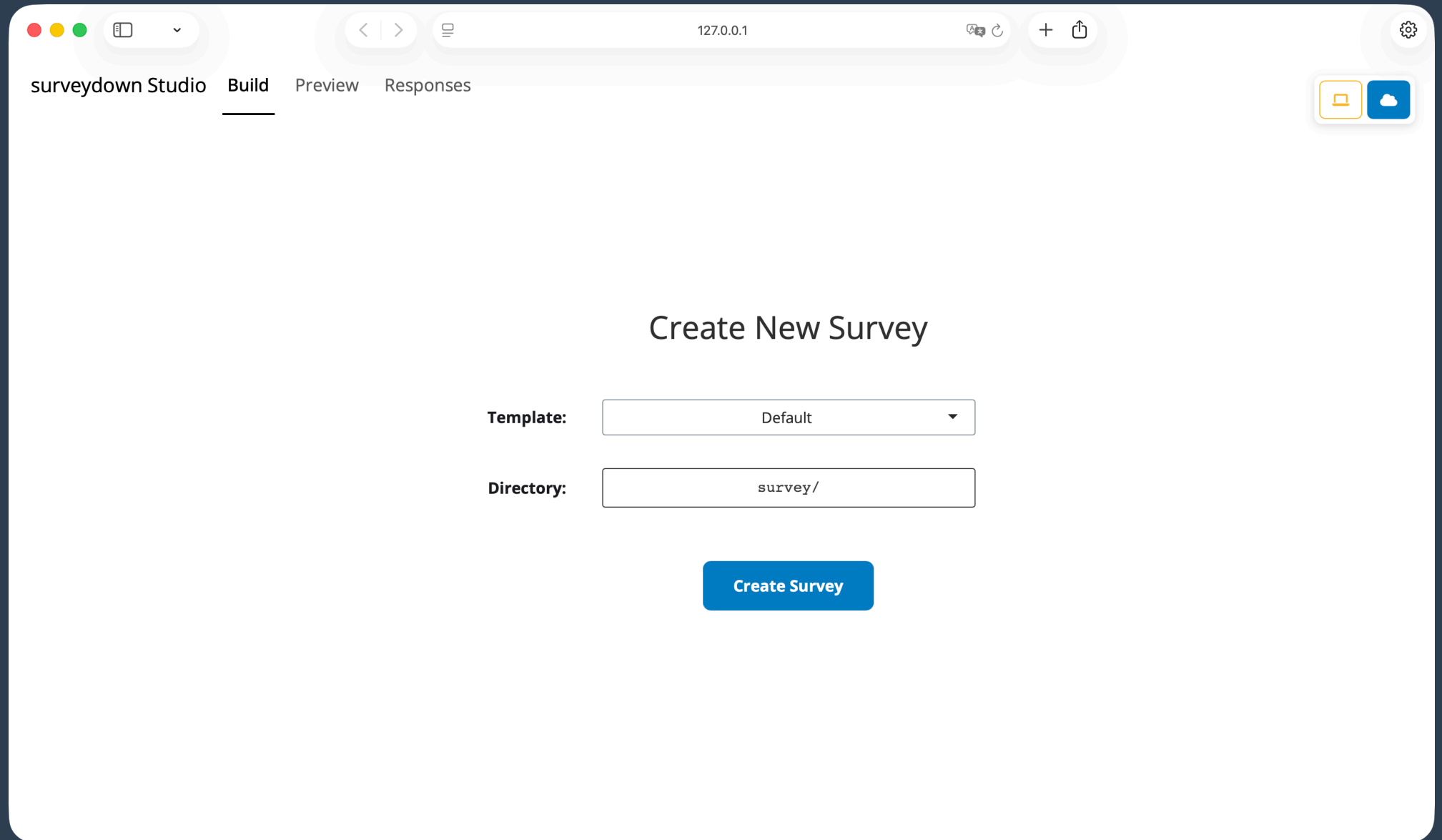
Studio Workflow



Launch with a simple command

```
sdstudio :: launch()
```

Landing Page



The screenshot shows a web browser window with the address bar displaying '127.0.0.1'. The browser's top bar includes standard navigation icons (back, forward, refresh) and a settings gear icon. Below the browser window, the 'surveydown Studio' interface is visible. It features a navigation menu with four tabs: 'surveydown Studio', 'Build' (which is currently selected and underlined), 'Preview', and 'Responses'. In the top right corner of the studio, there are two icons: a yellow one with a laptop and a blue one with a cloud. The main content area of the 'Build' tab is titled 'Create New Survey'. It contains two form fields: 'Template:' with a dropdown menu currently set to 'Default', and 'Directory:' with a text input field containing 'survey/'. Below these fields is a prominent blue button labeled 'Create Survey'.

surveydown Studio **Build** Preview Responses

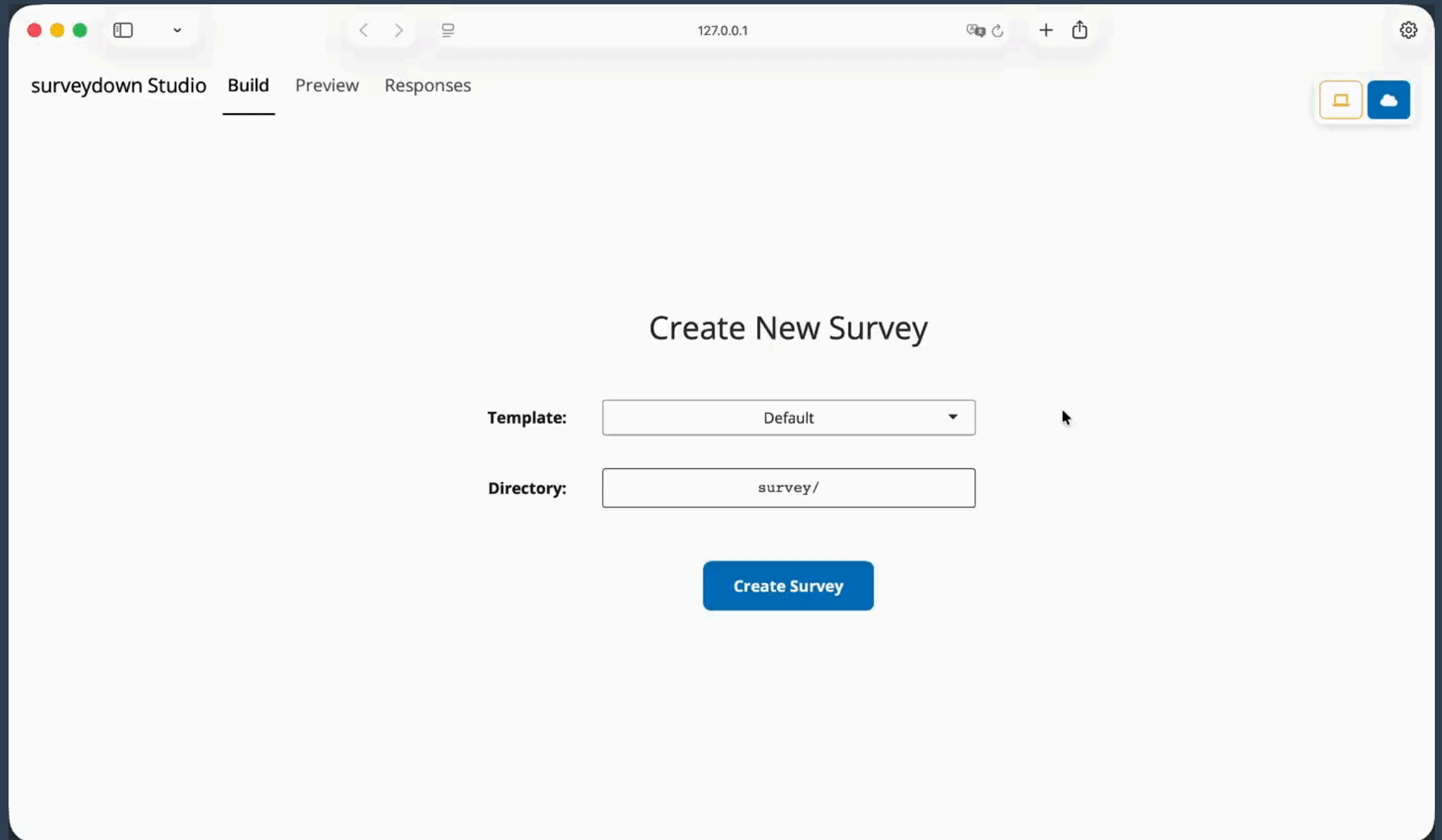
Create New Survey

Template: Default

Directory: survey/

Create Survey

Start with a Template



The screenshot shows the 'surveydown Studio' web application. The browser's address bar displays '127.0.0.1'. The application has a top navigation bar with tabs for 'Build' (active), 'Preview', and 'Responses'. On the right side of the top bar, there are icons for a folder, a cloud, and a settings gear. The main content area is titled 'Create New Survey'. It contains two form fields: 'Template:' with a dropdown menu set to 'Default', and 'Directory:' with a text input field containing 'survey/'. Below these fields is a blue button labeled 'Create Survey'.

surveydown Studio Build Preview Responses

Create New Survey

Template: Default

Directory: survey/

Create Survey

surveydown.org/templates

surveydown.org

About

Documentation

Templates

Blog

FAQ

1 surveydown Studio

Templates – surveydown.org

Templates

Basic

Default Template

Question Types

Questions Defined with YAML

Conditions

Conditional Showing

Conditional Skipping

Conditional Stopping

Randomization

Random Options

Random Options Predefined

Option Shuffling

Reactivity

Reactive Questions

Reactive drilldown

External Redirect

Live Polling

Conjoint

Conjoint Buttons

Conjoint Tables

Custom

Custom Leaflet Map

Custom Plotly Chart

Templates

To help you get started with surveydown, we created some templates that cover many of the features of the package. Feel free to use any as a starting point for your own surveys or as a reference for specific features.

Call `sd_create_survey()` with proper arguments to create your desired templates in your preferred directory. Refer to the [Start with a template](#) section for more details.

Default Template

Welcome to surveydown! This is a default template of a surveydown survey. It has two pages with one question on each page.
Here is a basic "multiple choice" question:

Which type of penguin do you like the best?

Adelie

Chinstrap

Gentoo

You need to insert next buttons with `sd_next()` on each page. By default, it will go to the next page defined in the `survey.yml` file, but you can also specify the page with the `next_page` argument set to the name of the page you want to go to next.

Source Code

Default Template

BASIC

A minimum template for starting from scratch.

Template - Question Types

Hi! This is a showcase of different question types and question formatting that surveydown supports. Please make your choice below:

Where do you want to go to?
Take me to the page on Question Types the next page
Take me to the page on Question Formatting
Skip to the end

Source Code

Question Types

BASIC

A template showcasing all built-in [question types](#).

Template - Questions Defined with YAML

This is a showcase of defining questions with a `questions.yml` file.
The `sd_questions()` also takes a `yml` argument default to `"questions.yml"`. If you have such file in your project root folder, you can use it to support your question definition.
You can also change to other file name or directory and define questions like this:

```
sd_questions(
  yml = "your_question.yml",
  yml = "folder_name/files_question.yml"
)
```


Please click on the "Next" button to see the demo questions.

Source Code

Questions Defined with YAML

BASIC

A template showcasing all built-in [question types](#) using a YAML file.

Categories

All (17)

Basic (3)

Conditions (3)

Conjoint (2)

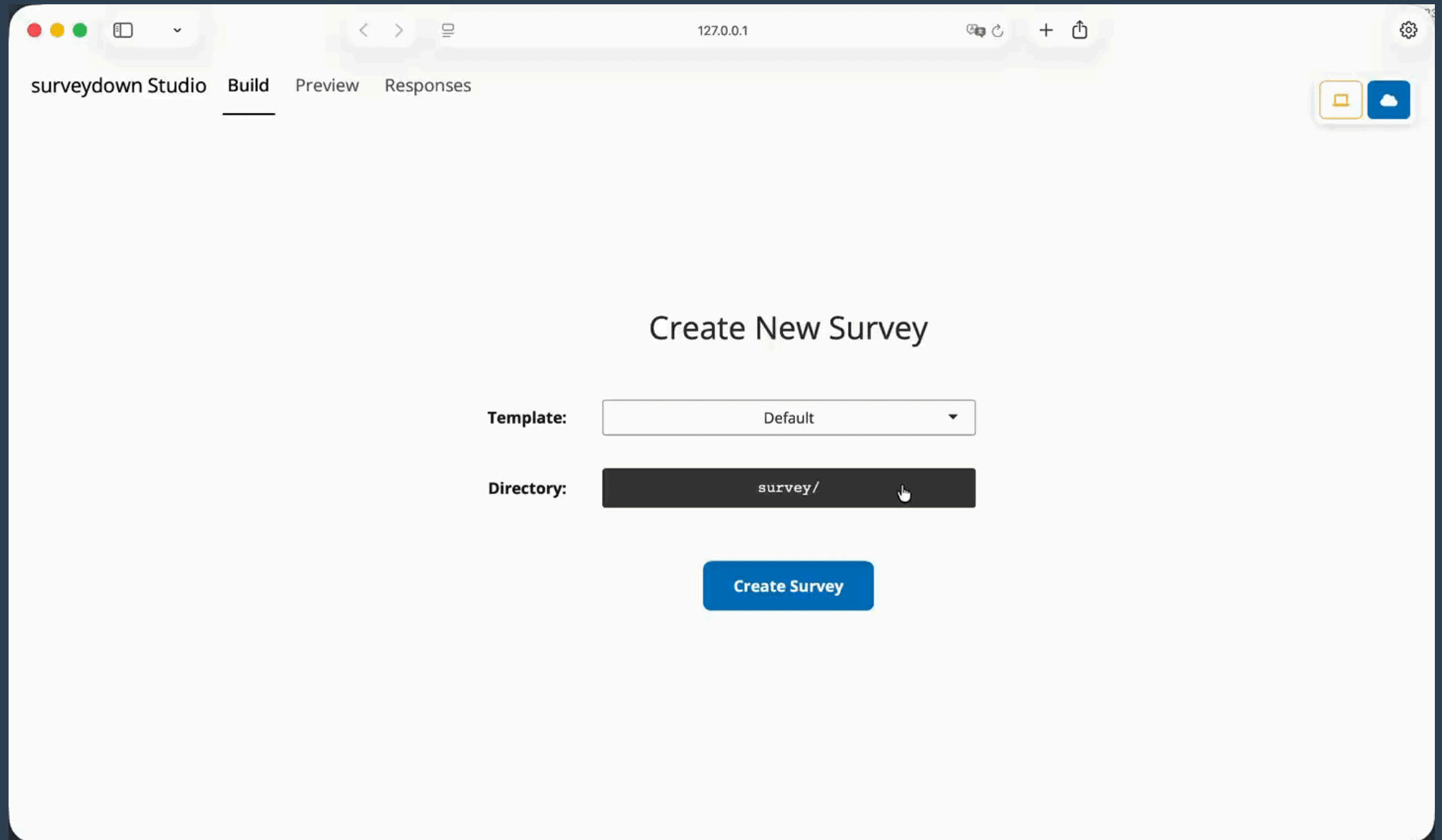
Custom (2)

Randomization (3)

Reactivity (4)

Site made with   and  quarto

Choose Your Directory



The screenshot shows the 'surveydown Studio' application interface. At the top, there is a navigation bar with tabs for 'Build', 'Preview', and 'Responses', with 'Build' being the active tab. The address bar shows '127.0.0.1'. The main content area is titled 'Create New Survey'. It features two input fields: 'Template:' with a dropdown menu set to 'Default', and 'Directory:' with a text input field containing 'survey/'. Below these fields is a blue button labeled 'Create Survey'.

surveydown Studio Build Preview Responses

127.0.0.1

Create New Survey

Template: Default

Directory: survey/

Create Survey

The Build Tab

surveydown Studio Build Preview Responses

Structure

- Page: welcome
 - # Default Template Welcome to...
 - penguins | mc
What's your favorite penguin?
 - There are several ways to...
- Page: page2
 - This is another page in...
 - silly_word | text
Write a silly word here:
 - On this page, we don't...
- Page: end
 - ## End This is the...

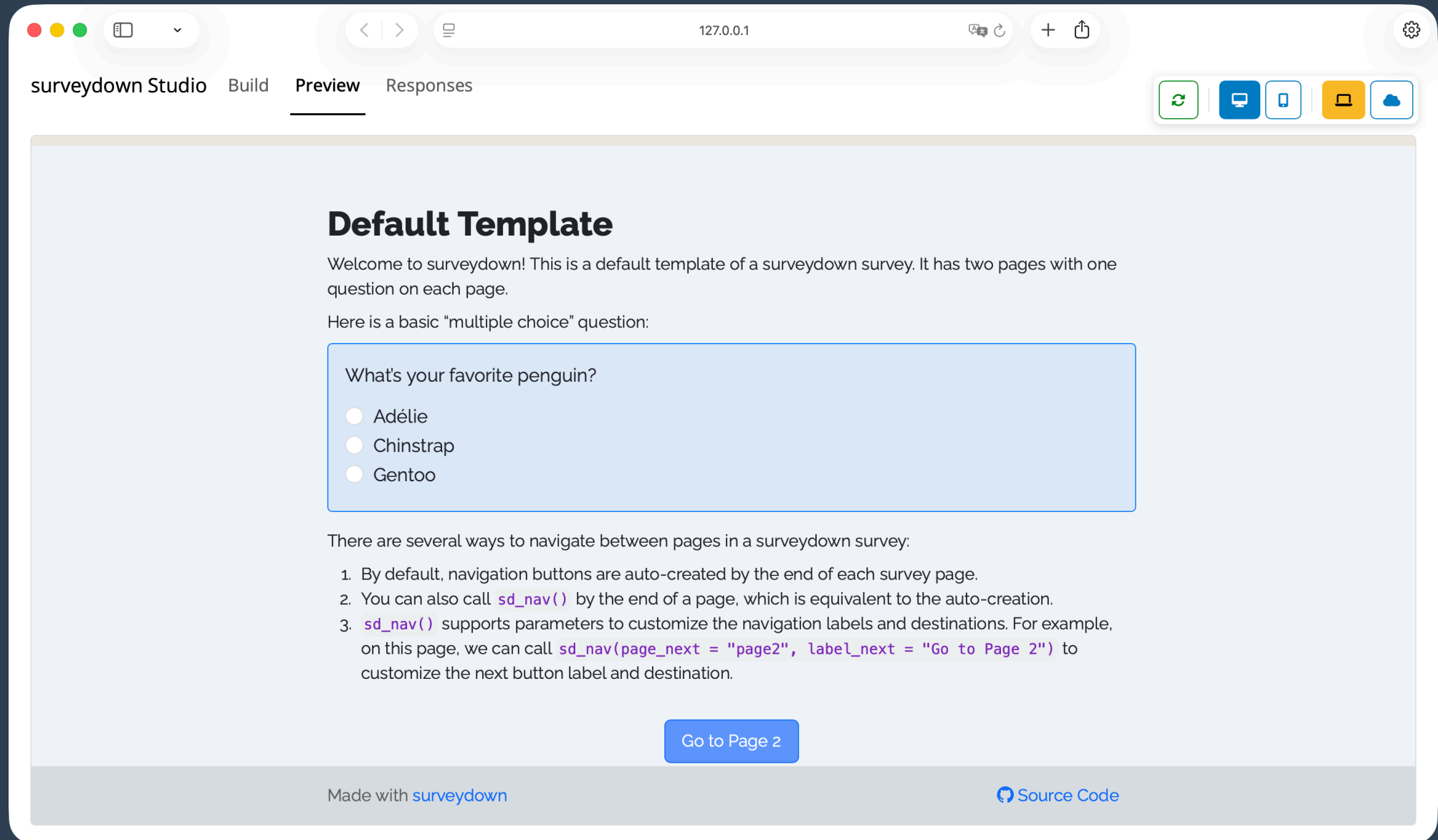
Add A New Page

Code

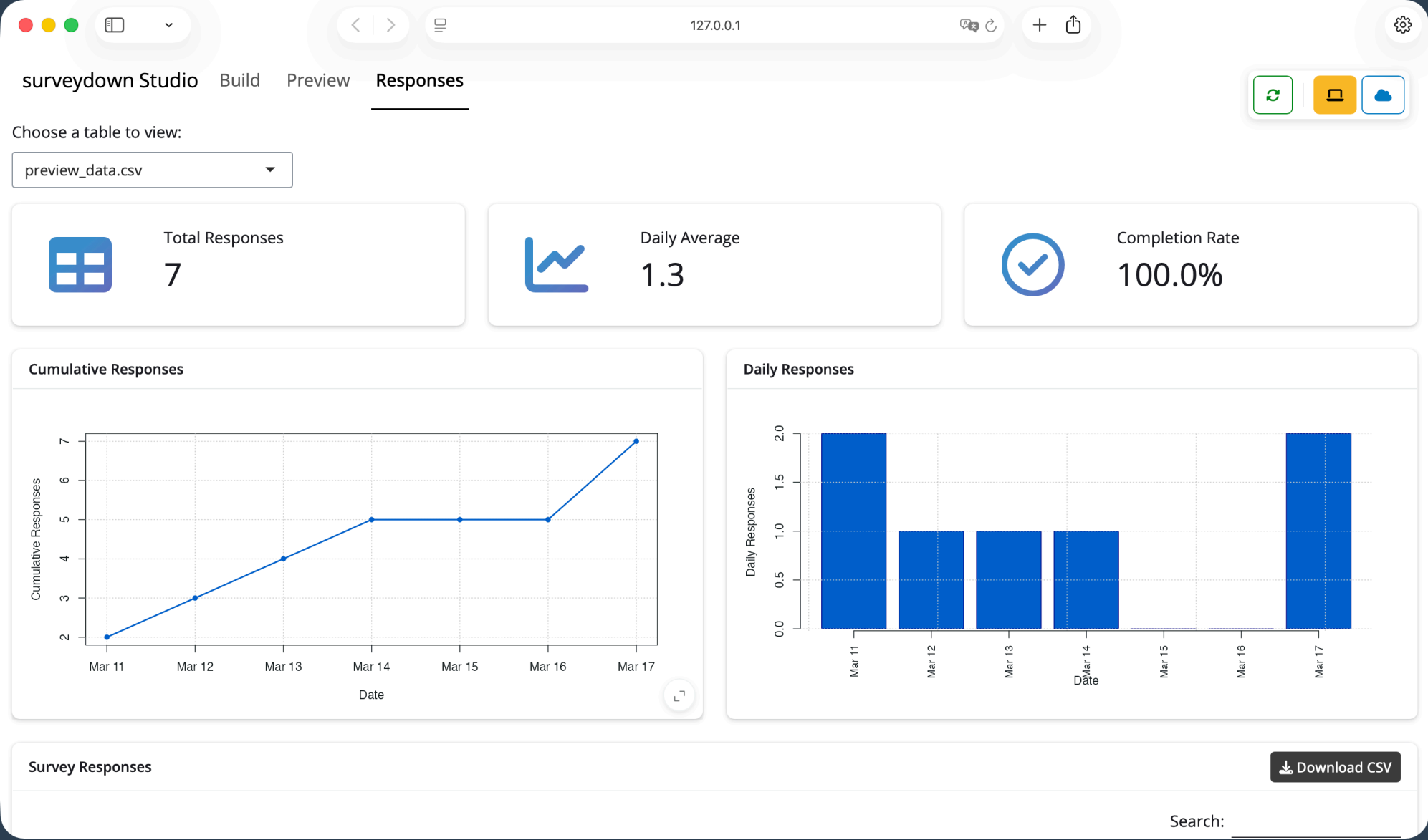
survey.qmd app.R

```
37 --- welcome
38
39 # Default Template
40
41 Welcome to surveydown! This is a default template of a surveydown survey. It has two
42 pages with one question on each page.
43
44 Here is a basic "multiple choice" question:
45
46 ```{r}
47 sd_question(
48   type = 'mc',
49   id   = 'penguins',
50   label = "What's your favorite penguin?",
51   option = c(
52     'Adélie' = 'adelie',
53     'Chinstrap' = 'chinstrap',
54     'Gentoo' = 'gentoo'
55   )
56 )
57
58 There are several ways to navigate between pages in a surveydown survey:
59
60 1. By default, navigation buttons are auto-created by the end of each survey page.
61 2. You can also call `sd_nav()` by the end of a page, which is equivalent to the
62    auto-creation.
63 3. `sd_nav()` supports parameters to customize the navigation labels and destinations.
64    For example, on this page, we can call `sd_nav(page_next = "page2", label_next = "Go to
65    Page 2")` to customize the next button label and destination.
66
67 ```{r}
```

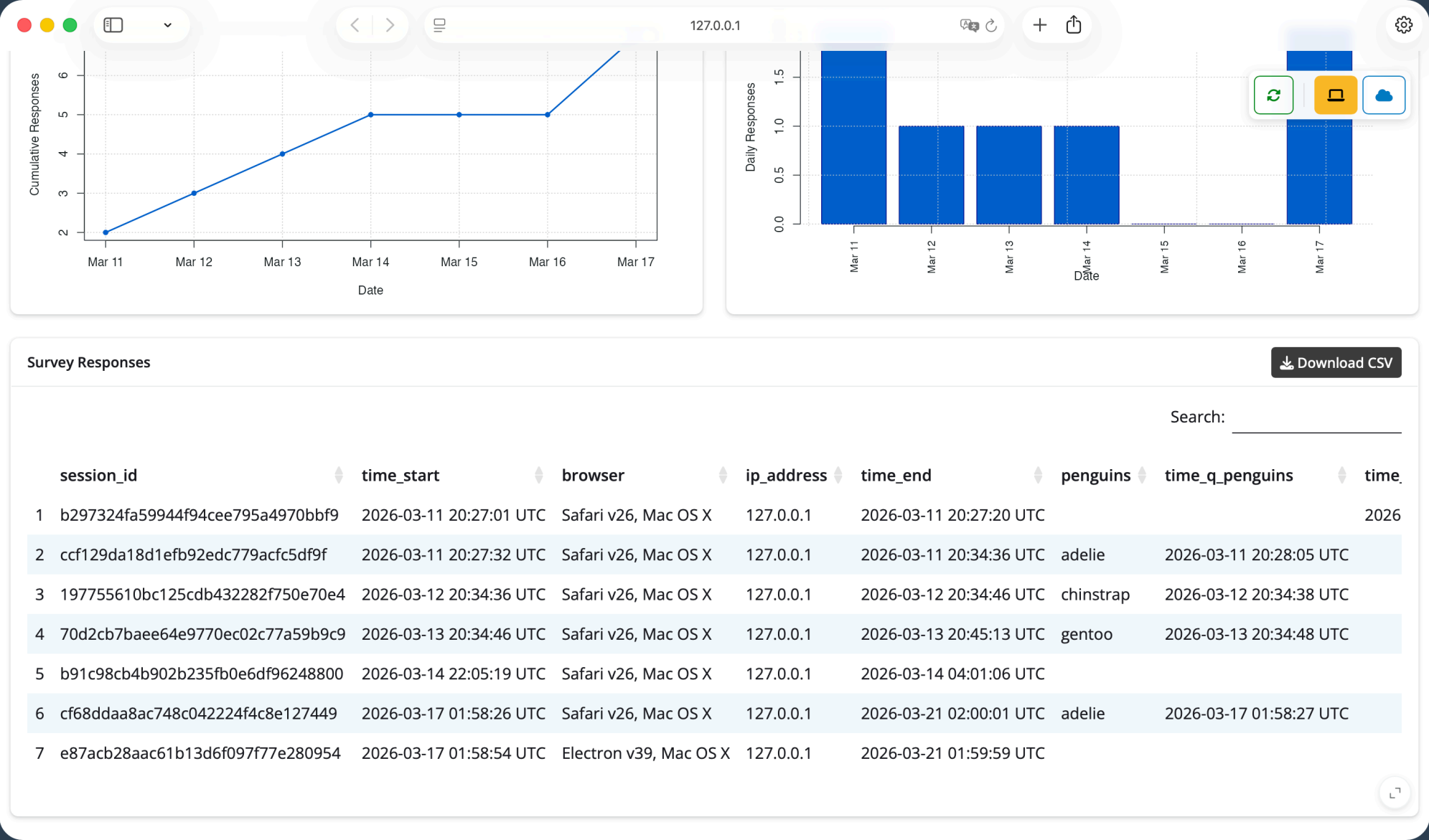
The Preview Tab



The Responses Tab

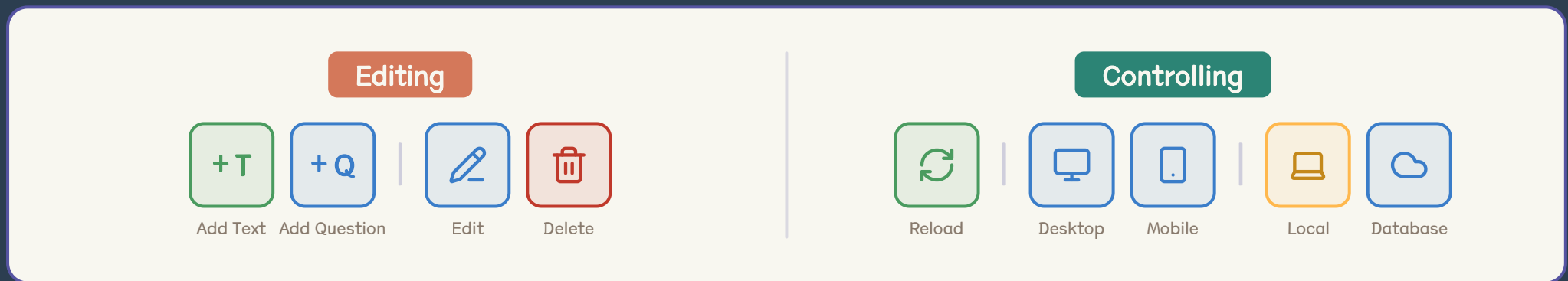


The Responses Tab - Continued



Studio Features

Self-explanatory Icons



Editing

GUI



Point & Click

No coding required

or

Script



Edit Directly

Full code control

GUI - Toggling

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- At the top, there are tabs for "surveydown Studio", "Build", "Preview", and "Responses".
- Below the tabs, there are icons for undo, redo, and a search icon.
- The main area shows a list of pages:
 - Page: page1** (with icons for +T, +Q, edit, delete, and a dropdown arrow):
 - A text block: `# Title Some texts.` (with edit and delete icons).
 - A question block: `penguins | mc` with the label `What's your favorite penguin?` (with edit and delete icons).
 - Page: end** (with icons for +T, +Q, edit, delete, and a dropdown arrow):
 - A text block: `## End This is the...` (with edit and delete icons).
- At the bottom, there is an orange button labeled "Add A New Page".

Code Panel:

- The top of the panel has tabs for "survey.qmd" and "app.R".
- The main area shows the R code for the survey application, with line numbers on the left:

```
33 library(surveydown)
34 ----
35
36
37 --- page1
38
39 # Title
40
41 Some texts.
42
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 ----
55
56 --- end
57
58 ## End
59
60 This is the last page in the survey.
61
62 ```{r}
63 sd_close()
64 ----
```

GUI - Content Editing

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel (Left):

- Page: page1**
 - Text: # Title Some texts.
 - Question: penguins | mc
What's your favorite penguin?
- Page: end**
 - Text: ## End This is the...

At the bottom of the Structure panel is a button labeled "Add A New Page".

Code Panel (Right):

The Code panel shows the R code for the survey, with line numbers 33 to 64. The code is as follows:

```
33 library(surveydown)
34 
35 
36 
37 --- page1
38 
39 # Title
40 
41 Some texts.
42 
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 
55 
56 --- end
57 
58 ## End
59 
60 This is the last page in the survey.
61 
62 ```{r}
63 sd_close()
64 
```

GUI - Content Addition

The screenshot displays the surveydown Studio interface, which is used for creating interactive surveys. The interface is divided into two main panels: Structure and Code.

Structure Panel (Left):

- Page: page1** (Selected)
 - # Title** New texts.
 - penguins | mc**
What's your favorite penguin?
- Page: end**
 - ## End** This is the...

Code Panel (Right):

The Code panel shows the R code for the survey, with line numbers 33 to 64. The code defines the survey structure and content for the selected page.

```
33 library(surveydown)
34 
35 
36 
37 --- page1
38 
39 # Title
40 
41 New texts.
42 
43 ```{r}
44 sd_question(
45   type = 'mc',
46   id = 'penguins',
47   label = "What's your favorite penguin?",
48   option = c(
49     'Adélie' = 'adelie',
50     'Chinstrap' = 'chinstrap',
51     'Gentoo' = 'gentoo'
52   )
53 )
54 
55 
56 --- end
57 
58 ## End
59 
60 This is the last page in the survey.
61 
62 ```{r}
63 sd_close()
64 
```

GUI - Page Addition

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- At the top, there are tabs for "surveydown Studio", "Build", "Preview", and "Responses".
- The "Structure" tab is active, showing a hierarchical view of the survey pages.
- Under "Page: page1", there are three items:
 - A text block labeled "# Title New texts." with a green edit icon and a red delete icon.
 - A multiple-choice question labeled "penguins | mc" with the text "What's your favorite penguin?". It has a green edit icon and a red delete icon.
 - A text question labeled "text_question | text" with the text "Write something:". It has a green edit icon and a red delete icon.
- Under "Page: end", there is one item:
 - A text block labeled "## End This is the..." with a green edit icon and a red delete icon.
- At the bottom of the Structure panel, there is an orange button labeled "Add A New Page".

Code Panel:

- The "Code" tab is active, showing the R code for the survey.
- The code is organized into two sections: "survey.qmd" and "app.R".
- The "survey.qmd" section contains the following code:

```
41 new texts.  
42  
43 ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54 ```  
55 ```{r}  
56 sd_question(  
57   type = "text",  
58   id   = "text_question",  
59   label = "Write something:"  
60 )  
61 ```
```
- The "app.R" section contains the following code:

```
62  
63  
64 --- end  
65  
66 ## End  
67  
68 This is the last page in the survey.  
69  
70 ```{r}  
71 sd_close()  
72 ```
```

GUI - Page Relocation

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel: This panel shows a hierarchical view of the survey pages. It includes three pages: "Page: page1", "Page: page2", and "Page: end". Each page has a list of questions or text blocks. For "Page: page1", there is a title "New texts.", a multiple-choice question "What's your favorite penguin?", and a text question "Write something:". For "Page: page2", there is a text block "This is page 2.". For "Page: end", there is a text block "## End This is the...". Each item in the list has edit and delete icons. At the bottom of the Structure panel, there is a button labeled "Add A New Page".

Code Panel: This panel shows the R code for the survey. It includes the following code:

```
41 new texts.  
42  
43 ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54 ```  
55 ```{r}  
56 sd_question(  
57   type = "text",  
58   id = "textquestion",  
59   label = "Write something:"  
60 )  
61 ```  
62  
63 --- page2  
64  
65 This is page 2.  
66  
67  
68 --- end  
69  
70  
71 ## End  
72
```

GUI - Content Relocation

The screenshot displays the surveydown Studio interface, which is divided into two main panels: Structure and Code.

Structure Panel:

- Page: page1**
 - # Title New texts.**
 - penguins | mc**
What's your favorite penguin?
 - text_question | text**
Write something:
- Page: page2**
 - This is page 2.**
- Page: end**
 - ## End This is the...**

At the bottom of the Structure panel is a button labeled "Add A New Page".

Code Panel:

The Code panel shows the R code for the survey, with tabs for "survey.qmd" and "app.R". The visible code is as follows:

```
41 new texts.  
42  
43 ```{r}  
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54 ```  
55 ```{r}  
56 sd_question(  
57   type = "text",  
58   id   = "text_question",  
59   label = "Write something:"  
60 )  
61 ```  
62  
63 --- page2  
64  
65 This is page 2.  
66  
67  
68 --- end  
69  
70  
71 ## End  
72
```

Script Editing

The screenshot displays the surveydown Studio interface, which is used for creating and editing surveys. The interface is divided into two main panels: Structure and Code.

Structure Panel: This panel shows the survey's structure, including pages and questions. It has tabs for "Build", "Preview", and "Responses". The "Build" tab is active. The structure is as follows:

- Page: page1**
 - Text question: "# Title New texts."
 - Multiple choice question: "penguins | mc" with the label "What's your favorite penguin?"
- Page: page2**
 - Text question: "This is page 2."
 - Text question: "text_question | text" with the label "Write something:"
- Page: end**
 - Text question: "## End This is the..."

At the bottom of the Structure panel is a button labeled "Add A New Page".

Code Panel: This panel shows the R script used to generate the survey. The script is named "survey.qmd" and "app.R". The code is as follows:

```
44 sd_question(  
45   type = 'mc',  
46   id   = 'penguins',  
47   label = "What's your favorite penguin?",  
48   option = c(  
49     'Adélie' = 'adelie',  
50     'Chinstrap' = 'chinstrap',  
51     'Gentoo' = 'gentoo'  
52   )  
53 )  
54 ~~~  
55 --- page2  
56  
57 This is page 2.  
58  
59 ~~~{r}  
60 sd_question(  
61   type = "text",  
62   id   = "text_question",  
63   label = "Write something:"  
64 )  
65 ~~~  
66  
67 --- end  
68  
69 ## End  
70  
71 This is the last page in the survey.  
72  
73 ~~~{r}  
74 sd_close()  
75 ~~~
```

Instant Preview

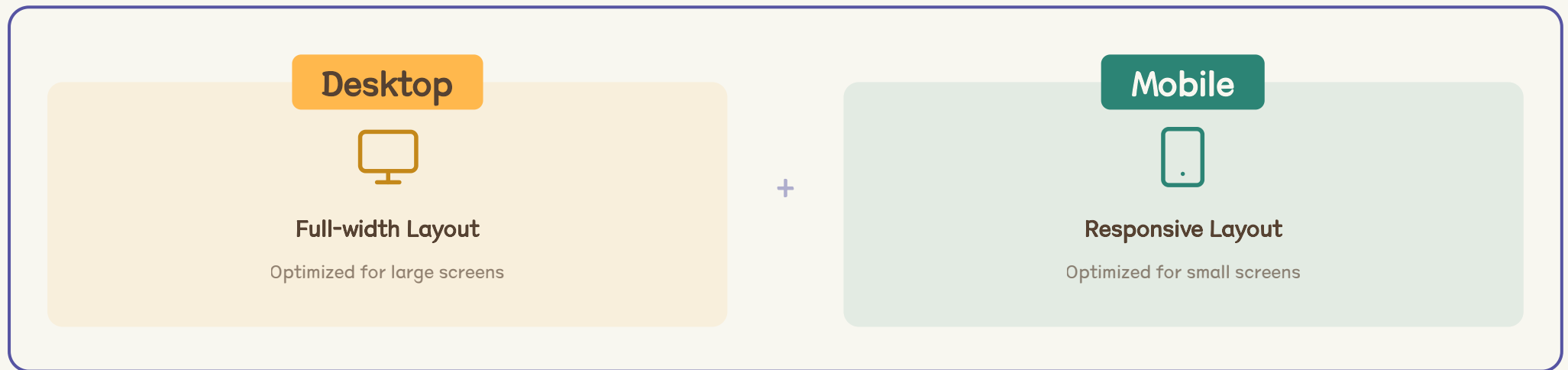
The screenshot displays the surveydown Studio application interface. At the top, there's a navigation bar with tabs for 'surveydown Studio', 'Build', 'Preview', and 'Responses'. The 'Build' tab is currently active. Below the navigation bar, the interface is split into two main panels: 'Structure' on the left and 'Code' on the right.

Structure Panel: This panel shows a hierarchical view of the survey structure. It lists three pages: 'Page: page1', 'Page: page2', and 'Page: end'. Each page has associated elements: 'Page: page1' contains a title '# Title New texts.' and a multiple-choice question 'penguins | mc' with the label 'What's your favorite penguin?'. 'Page: page2' contains a text element 'This is page 2.' and a text question 'text_question | text' with the label 'Write something:'. 'Page: end' contains a title '## End This is the...'. Each element has icons for adding, editing, and deleting.

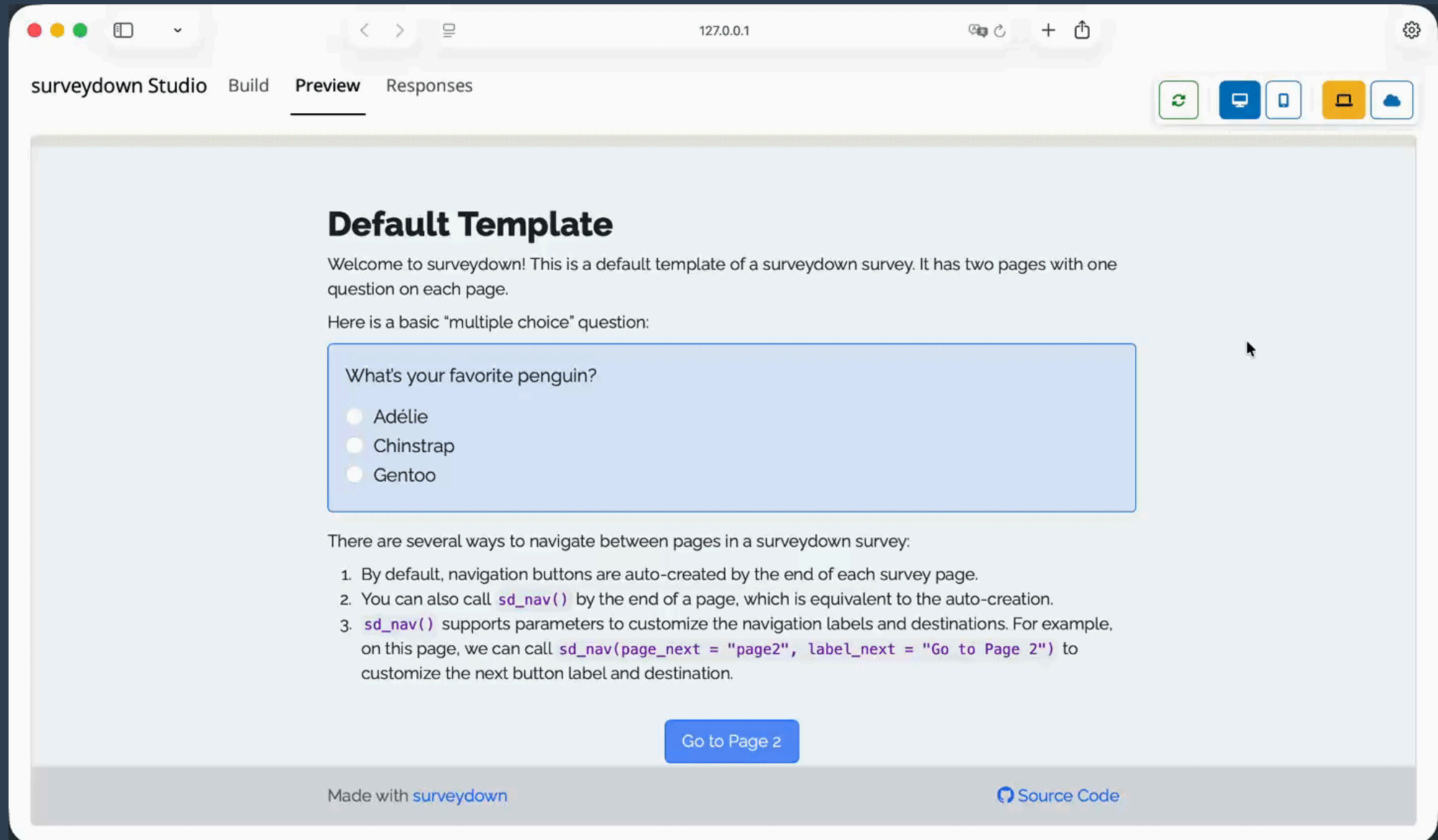
Code Panel: This panel shows the R code for the survey. It includes a file explorer at the top with 'survey.qmd' and 'app.R'. The code is as follows:

```
34- ...
35- ...
36- ...
37- --- page1
38- # Title
39- New texts.
40- ...{r}
41- sd_question(
42-   type = 'mc',
43-   id = 'penguins',
44-   label = "What's your favorite penguin?",
45-   option = c(
46-     'Adélie' = 'adelie',
47-     'Chinstrap' = 'chinstrap',
48-     'Gentoo' = 'gentoo'
49-   )
50- )
51- ...
52- --- page2
53- This is page 2.
54- ...{r}
55- sd_question(
56-   type = "text",
57-   id = "text_question",
58-   label = "Write something:"
59- )
60- ...
```

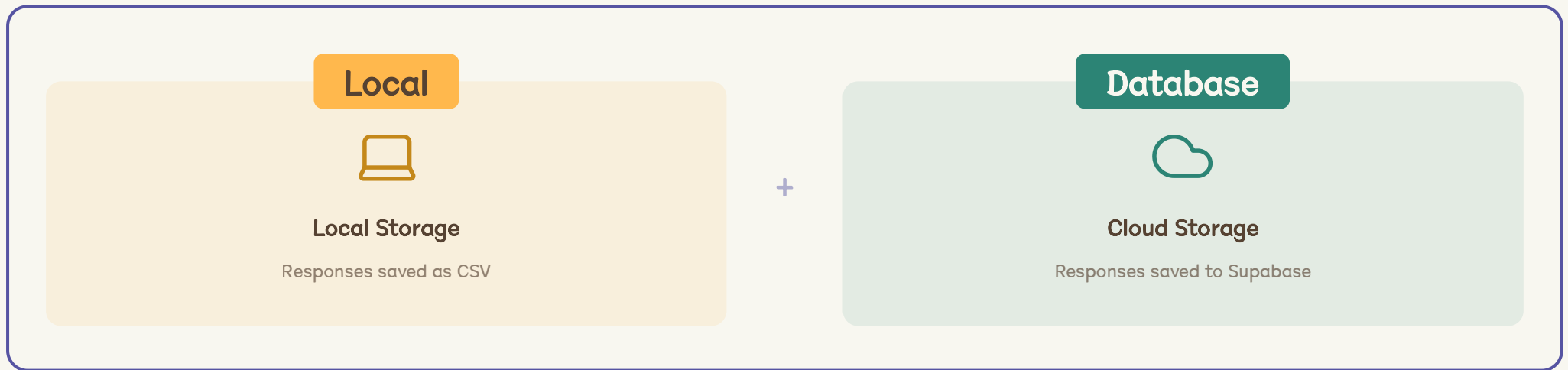
Views



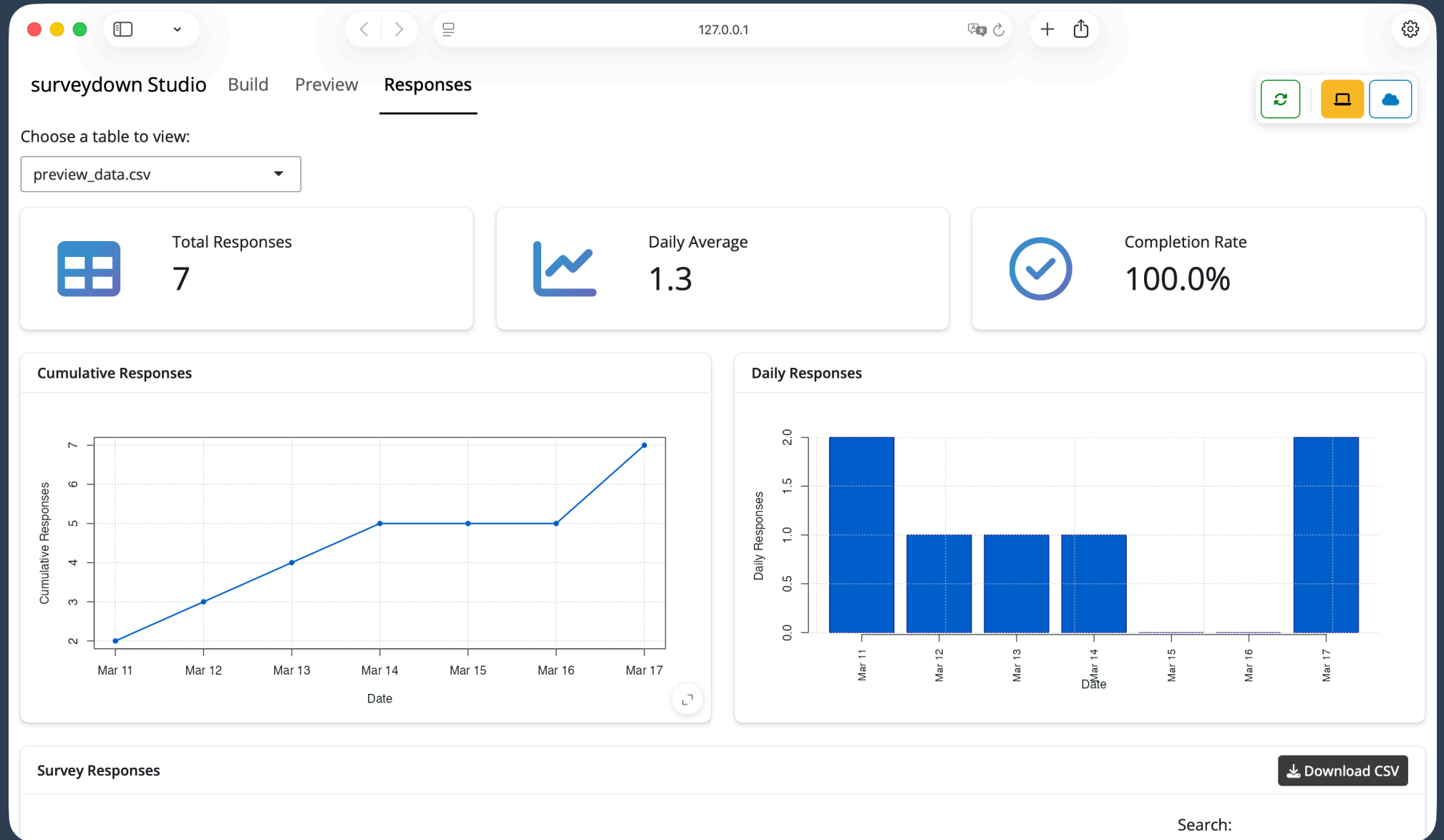
Desktop and Mobile Views



Responses



Local Responses



Database Responses

127.0.0.1

surveydown Studio Build Preview Responses

Database Connection [Settings](#)

Not connected

Choose a table to view:

Select or type table name... ▼

Total Responses

Daily Average

Completion Rate

Cumulative Responses

Daily Responses

Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Settings

Not connected

Choose a table to view:

Select or type table name...

Total Responses

Daily Average

Completion Rate

Cumulative Responses

Daily Responses

Database Responses

surveydown Studio

Build

Preview

Responses

Database Connection

Close

Not connected

Host:

Port:

Database Name:

postgres

User:

Password:

.....

Show

Table:

default

Test Connection

Choose a table to view:

Select or type table name...

Database Responses

surveydown Studio

Build

Preview

Responses

127.0.0.1

Connected (GSSAPI disabled)

Database Connection

Close

Host:

Port:

Database Name:

postgres

User:

Password:

.....

Show

Table:

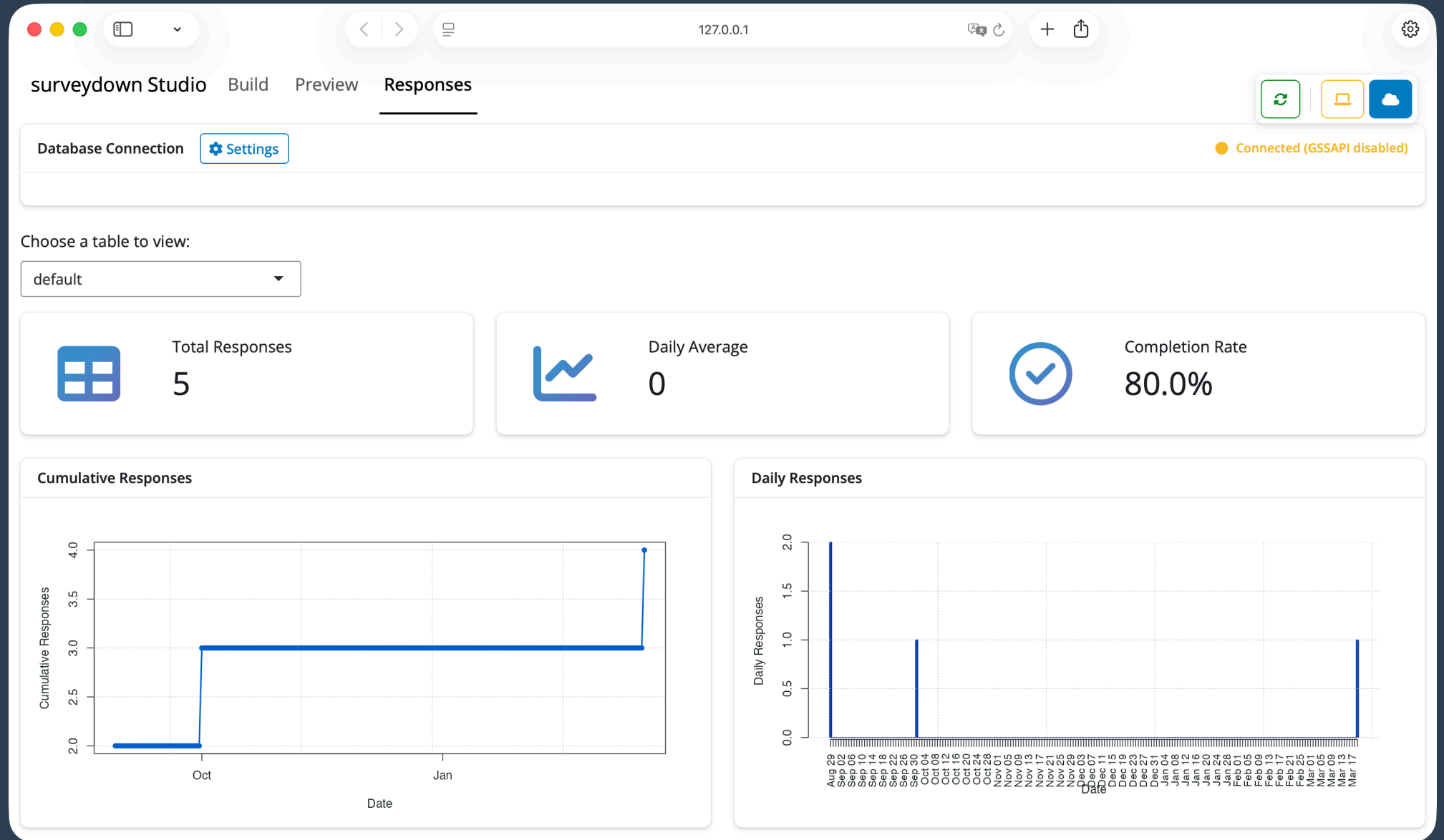
default

Test Connection

Choose a table to view:

default

Database Responses



sdstudio gives you...



GUI for Survey Editing



Live Preview



Response Dashboard



SQL Connection



Data View & Download



...and more!

surveydown + sdstudio

surveydown

 Text-based editing

 Published paper

 8000+ CRAN downloads

sdstudio

 GUI editing & live preview

 Actively developed

Open to contributions

 Better together!

Ways to help:

github.com/surveydown-dev/sdstudio/

- Try it out!
- Give it a ★
- Post an issue
- Join the GitHub discussion
- Contribute new features



Meet the team!

surveydown.org



John Helveston, Ph.D.



Pingfan Hu



Bogdan Bunea

